BUS 20100: Financial Accounting

DNA Storage, Inc. Solution

DNA Storage, Inc.: Incorporation

1. 7,000,000 shares are issued to investors for $5 per share.
2. Additional 1,000,000 shares were given to the founders in exchange for the trademark "DNAStorage".

[in $1,000]

Cash (A)

BB	0
(1)	35,000

7 million shares × $5

Trademark (A)

BB	0
(2)	5,000

Common Stock – Paid in Capital (SE)

0	BB
35,000	(1)
5,000	(2)

DR Cash (+A) 35,000
DR Trademark (+A) 5,000
CR Common stock – Paid in Capital (+SE) 40,000

DNA Storage, Inc.

3. On January 15, 2017 DS Inc. signed a one year licensing agreement with the research laboratory that owns a patent for retrieving data from synthetic DNA molecules. The licensing period begins February 1, 2017 and the annual fee of $6,000,000 was paid in full.

Conceptual issue: Capitalize or Expense?

Capitalize or Expense?

- **To make this determination, we appeal to the definition of an asset:**

  Asset – an economic **resource** with probable **future benefit**, that can be **measured** with a **reasonable** degree of precision and is **owned** by the firm.

- **Does this expenditure create an asset?**

  - If yes, then **capitalize** (i.e., record as an asset right now), and expense it over future periods (as the asset "expires/is used up").
  - If no, then **expense** immediately (i.e., decrease shareholders' equity immediately).

DNA Storage, Inc.

3. On January 15, 2017 DS Inc. signed a one year licensing agreement with the research laboratory. The licensing period begins February 1, 2017 and the annual fee of $6,000,000 was paid in full.

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)

Prepaid License (A)			
BB	0		
(3)	6,000		

DR Prepaid License (+A) 6,000

CR Cash (-A) 6,000

DNA Storage, Inc.

4. DS Inc. contracted with the small biotechnology company to design an efficient technology of transcribing digital data into the molecule. Towards the end of February 2017, the biotechnology company delivered the required specifications to DS Inc. and was paid a total of $2,000,000 in cash on the same day.

Conceptual issue: Capitalize or Expense?

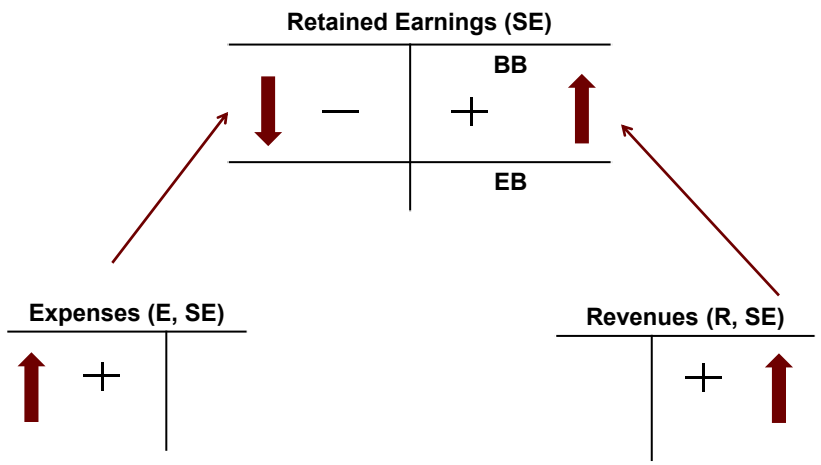
Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)

R&D Expense (E, SE)			
(4)	2,000		

DR R&D Expense (+E, -SE) 2,000

CR Cash (-A) 2,000

T-accounts for Revenues and Expenses



DNA Storage, Inc.

5. On February 15, 2017 DS Inc. purchased a machine used in the production of DNA-based storage systems for $6,000,000 in cash.

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)

Machinery (A)			
BB	0		
(5)	6,000		

DR Machinery (+A) 6,000
CR Cash (-A) 6,000

DNA Storage, Inc.

6. On February 25, 2017, DS paid $7,000,000 upon receiving a delivery of materials for the use in the production of DNA-based storage systems.

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)

Materials (A)			
BB	0		
(6)	7,000		

DR Materials (+A) 7,000
CR Cash (-A) 7,000

DNA Storage, Inc.

7. On July 1, 2017 DS Inc. paid $600,000 for salaries of corporate officers and employees (for the January-June period).

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)

Salaries (E, SE)			
(7)	600		

DR Salaries (+E, -SE) 600
CR Cash (-A) 600

DNA Storage, Inc.

8. On July 1, 2017 borrowed $7,000,000 from a bank for the annual interest rate of 15% which is paid semi-annually.

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		

Bank Loan (L)			
		0	BB
		7,000	(8)

DR Cash (+A) 7,000
CR Bank Loan (+L) 7,000

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DNA Storage, Inc.

9. Additional materials for the production of DNA-based storage systems were purchased for the total of $4,000,000. Under the terms of suppliers, DS Inc. will make no payments until February 2018.

Materials (A)			
BB	0		
(6)	7,000		
(9)	4,000		

Accounts Payable (L)			
		0	BB
		4,000	(9)

DR Materials (+A) 4,000
CR Accounts Payable (+L) 4,000

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DNA Storage, Inc.

10. During the six months ended December 31, 2017, the company paid $3,000,000 for insurance (for the July-December period) and other general expenses such as utilities.

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)

General Expenses (E,SE)			
(10)	3,000		

DR General Expenses (+E, - SE) 3,000

CR Cash (+A) 3,000

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DNA Storage, Inc.

11. An additional $1,400,000 in cash was paid for corporate salaries and other corporate expenses (for the July-December period).

Cash (A)			
BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)
		1,400	(11)

Salaries (E,SE)			
(7)	600		
(11)	1,400		

DR Salaries (+E, - SE) 1,400

CR Cash (-A) 1,400

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DNA Storage, Inc.

12. DS Inc. spend $500,000 in cash on advertising in 2017. The advertising was mostly done through scientific popular journals, marketing materials and presentations at biotech conferences.

Conceptual issue: Capitalize or expense?

Cash (A)

BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)
		1,400	(11)
		500	(12)

Advertising Expense (E,SE)

(12)	500		

DR Advertising Expense (+E, -SE) 500

CR Cash (-A) 500

DNA Storage, Inc.

13. Towards the end of 2017, orders for DNA-based storage systems started to arrive. DS Inc. completed and shipped a total of $23,000,000 of its orders. As of the end of the year, a total of $3,500,000 of these sales had still not yet been collected.

Conceptual issues:

- **Revenue recognition**
- **Cost of goods sold**
  - **COGS (E, SE) ↑ by ?**
  - **Materials (A) ↓ by ?**

Recognition of Revenues and Expenses

- **In which period to record (i.e. recognize) revenues in financial statements?**
- **When revenues are both:**
  - **Earned** - the goods are delivered or a service is performed.
  - **Realized** - cash or a claim to cash (credit) is received in exchange for the goods or services.
- **In which period to record (i.e. recognize) expenses in financial statements?**
- **According to the matching principle:**
  - **Product costs (easily traceable to the revenue)** - in the same period when related revenues are recognized (i.e. cost of goods sold).
  - **Period costs (not easily traceable to the revenue)** - in the same period when these costs produce benefits (i.e. rent).

DNA Storage, Inc.

13. Towards the end of 2017, orders for DNA-based storage systems started to arrive. DS Inc. completed and shipped a total of $23,000,000 of its orders. As of the end of the year, a total of $3,500,000 of these sales had still not yet been collected.

Cash (A)

BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)
		1,400	(11)
		500	(12)
(13)	19,500		

Accounts Receivable (A)

BB	0		
(13)	3,500		

Revenue (R,SE)

		23,000	(13)

DR Cash (+A) 19,500
DR Accounts Receivable (+A) 3,500
CR Revenue (+R, +SE) 23,000

DR Cost of Goods Sold (+E, -SE) XXX
CR Materials (-A) XXX

DNA Storage, Inc.: First year of operations

14. On December 15, 2017, DS Inc. signed a contract with another research-oriented firm for a delivery of a sample DNA-based storage system. The selling price specified in a contract is $500,000. DS Inc. promised to deliver the machine on March 15, 2018. A research company has up to 30 days after the delivery to pay.

Conceptual issues:

- **Have we earned any revenue?**
- **Is this a liability?**

This is a contractual commitment.

Not recognized in the financial statements, but if it is material, it should be **disclosed** in the footnotes.

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DNA Storage, Inc.

15. At the end of the year DS Inc. paid $525,000 in interest on their bank loan.

Cash (A)

BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)
		1,400	(11)
(13)	19,500	500	(12)
		525	(15)

Interest Expense (E,SE)

(15)	525		

DR Interest Expense (+E, -SE) 525
CR Cash (-A) 525

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DNA Storage, Inc.

16. DS Inc. received $400,000 in interest on its cash account.

Cash (A)

BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)
		1,400	(11)
		500	(12)
(13)	19,500		
		525	(15)
(16)	400		

Interest Income (R,SE)

		400	(16)

DR Cash (+A) 400

CR Interest Income (+R, +SE) 400

End-of-period adjusting entries

- **Explicit Transactions:** Most transactions are triggered by observable events such as cash receipts, cash payments, credit sales, etc.
- **Implicit Transactions:** But some economic events **occur through the passage of time**, and do not generate source documents.
  - Two basic types of implicit transactions:
    - Incurring expenses as time passes.
    - Earning revenues as time passes.
  - Such events are not recognized in the accounting records until the end of an accounting period.
- **Accrual adjustments (also called "Adjusting Entries"):** End-of-period entries that record the financial effects of implicit transactions.

Adjusting entries

Passing of time and the **continuous use** of services/goods → Expense in the Income Statement and Increase in Liabilities or Decrease in Assets on the Balance Sheet

Example:

DR Depreciation Expense (+E, -SE) XXX
CR PP&E (-A) XXX

Passing of time and the **continuous rendering** of services/goods → Revenues in the Income Statement and Decrease in Liabilities or Increase in Assets on the Balance Sheet

Example:

DR Interest Receivable (+A) XXX
CR Interest Revenue (+R, +SE) XXX

DNA Storage, Inc.: Year-end adjustments

17. The licensing agreement with the research laboratory is due to expire on January 31, 2018. As of December 31, 2017, DS Inc. was still negotiating the terms of an extension of the licensing agreement.

Prepaid License (A)

BB	0		
(3)	6,000		
		5,500	(17)
EB	500		

= 6,000 * (11 / 12)

License Fees (E,SE)

(17)	5,500		

DR License Fees (+E, -SE) 5,500
CR Prepaid License (-A) 5,500

DNA Storage, Inc.: Year-end adjustments

18. As of December 31, 2017 there was still $2,300,000 worth of materials. There were no finished or partially finished storage systems.

Materials (A)		
BB	0	
(6)	7,000	
(9)	4,000	
		Materials used ?
EB	2,300	

Materials used (E,SE)		

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DNA Storage, Inc.: Year-end adjustments

18. As of December 31, 2017 there was still $2,300,000 worth of materials. There were no finished or partially finished storage systems.

Materials (A)		
BB	0	
(6)	7,000	
(9)	4,000	
		8,700 PLUG(18)
EB	2,300	

Materials used (E,SE)		
(18)	8,700	

= 7,000 + 4,000 – 2,300

DR Materials used (+E, -SE) 8,700

CR Materials (-A) 8,700

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DNA Storage, Inc.: Year-end adjustments

19. The machine used in the production of DNA-based storage systems is expected to last for approximately five years (one year of which has already passed). DS Inc. does not expect the machine to have any remaining value after five years.

Machinery (A)			
BB	0		
(5)	6,000		
		1,200	(19)
EB	4,800		

= 6,000/5

Depreciation Expense (E,SE)		
(19)	1,200	

DR Depreciation Expense (+E, -SE) 1,200

CR Machinery (-A) 1,200

DNA Storage, Inc.: Year-end adjustments

20. The trademark "DNAStorage" is initially valid for 10 years and may be renewed indefinitely for 10-year periods. The filing fee for a trademark renewal is $400. A valuation analysis at the end of the year suggests that the value of the trademark has increased to about $7,000,000.

Conceptual issues:

- **Economic life versus legal life? -> Economic**
- **Historic cost versus "market value"? -> Historic**

This trademark is a purchased asset, with an **indefinite economic life**. Under GAAP: We record it at its purchase price. We do not depreciate it, and we do not write it up for any increases in value. However, if there is evidence that its carry value is impaired, we need to write it down.

DNA Storage, Inc.: Assets at December 31, 2017

Cash (A)

BB	0		
(1)	35,000		
		6,000	(3)
		2,000	(4)
		6,000	(5)
		7,000	(6)
		600	(7)
(8)	7,000		
		3,000	(10)
		1,400	(11)
(13)	19,500	500	(12)
		525	(15)
(16)	400		
EB	34,875		

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DNA Storage, Inc.: Assets at December 31, 2017 (continued)

Trademark (A)

BB	0		
(2)	5,000		
EB	5,000		

Prepaid License (A)

BB	0		
(3)	6,000		
		5,500	(17)
EB	500		

Materials (A)

BB	0		
(6)	7,000		
(9)	4,000		
		8,700	(18)
EB	2,300		

Machinery (A)

BB	0		
(5)	6,000		
		1,200	(19)
EB	4,800		

Accounts Receivable (A)

BB	0		
(13)	3,500		
EB	3,500		

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DNA Storage, Inc.: Liabilities and Shareholders' equity at December 31, 2017

Bank Loan (L)		
	0	BB
	7,000	(8)
	7,000	EB

Accounts Payable (L)		
	0	BB
	4,000	(9)
	4,000	EB

Common Stock Paid in Capital (SE)		
	0	BB
	35,000	(1)
	5,000	(2)
	40,000	EB

Retained Earnings?

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DNA Storage, Inc.: Income Statement for 2017

[**in** $1,000]

Sales Revenues		**$23,000**
Expenses		
Materials used	8,700	
Advertising	500	
General Expenses	3,000	
Salaries	2,000	
R&D Expense	2,000	
License Fees	5,500	
Depreciation	1,200	
Interest expense net of income	125	
		(23,025)
Net Income (Loss)		**$(25)**

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DNA Storage, Inc.: Balance Sheet at December 31, 2017

[in $1,000]

Assets		Liabilities and Owners' Equity		
Cash	$34,875	Accounts Payable	$4,000	
Receivables	3,500	Bank Loan	7,000	
Materials	2,300	**Total Liabilities**		**11,000**
Prepaid License Fees	500	Common Stock	40,000	
Machinery, net	4,800	Retained Earnings	(25)	
Trademark	5,000	**Total Owners' Equity**		**39,975**
Total Assets	**$50,975**	**Total Liabilities and Owners Equity**		**$50,975**

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DNA Storage, Inc.: Statement of Cash Flows 2017
Begin by classifying all the cash transactions

Cash (A)

CFF	Paid-in-Capital	(1)	35,000				
				6,000	(3)	Licensing Fee	CFO
				2,000	(4)	R&D Expense	CFO
				6,000	(5)	Acq. of Machinery	CFI
				7,000	(6)	Purchase Materials	CFO
				600	(7)	Salaries	CFO
	Bank Loan	(8)	7,000				
				3,000	(10)	General Expenses	CFO
				1,400	(11)	Salaries	CFO
				500	(12)	Advertising	CFO
CFO	Cash collection from customers	(13)	19,500				
				525	(15)	Interest payment	CFO
	Interest income	(16)	400				
		EB	34,875				

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DNA Storage, Inc.: Statement of Cash Flows 2017 Direct method

Cash from Operations:	
Cash received from:	
Customers	$19,500
Interest collections	400
Cash paid for:	
General Expenses	(3,000)
Salaries	(2,000)
Advertising	(500)
Materials	(7,000)
Interest payments	(525)
Research & development	(2,000)
Licensing Fee	(6,000)
Net Cash from (used in) Operations	**$(1,125)** ← CFO

Direct method: simply group all the operating cash flow (CFO) items as either cash received, or cash paid for

The difference will be the net cash flow from operations (CFO)

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DNA Storage, Inc.: Statement of Cash Flows 2017 (continue)

Cash from Investing:		
Acquisition of machinery	$(6,000)	CFI
Net cash from (used in) investing	**$(6,000)**	
Cash from Financing:		
Issuance of common stock	$35,000	CFF
Bank borrowing	7,000	
Net cash from (used in) financing	**$42,000**	
Net Cash Flows	**$34,875**	← CFO+CFI+CFF
Cash balance at the beginning of the year	**0**	
Cash balance at the end of the year	**$34,875**	

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DNA Storage, Inc.: Income Statement for 2017

[in $1,000]

Sales Revenues			**$23,000**
Expenses			
Materials used	8,700		
Advertising	500		
General Expenses	3,000		
Salaries	2,000		
R&D Expense	2,000		
License Fees	5,500		
Depreciation	**1,200**	←	**Non-cash Expense**
Interest expense net of income	125		
			(23,025)
Net Income (Loss)			**$(25)**

Step 1: What are the non-cash revenues and expenses?

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DNA Storage, Inc.: Balance Sheet at December 31, 2017

[in $1,000]

Assets		**Liabilities and Owners' Equity**		
Cash	$34,875	Accounts Payable	$4,000	
Receivables	3,500	Bank Loan	7,000	
Materials	2,300	**Total Liabilities**		**11,000**
Prepaid License Fees	500	Common Stock	40,000	
Machinery, net	4,800	Retained Earnings	(25)	
Trademark	5,000	**Total Owners' Equity**		**39,975**
Total Assets	**$50,975**	**Total Liabilities and Owners Equity**		**$50,975**

Step 2: What are the B/S changes in non-cash current assets and liabilities?

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DNA Storage, Inc.: Balance Sheet at December 31, 2017

[in $1,000]

Assets		Liabilities and Owners' Equity		
Cash	$34,875	**Accounts Payable**	**$4,000**	
Receivables	**3,500**	Bank Loan	7,000	
Materials	**2,300**	**Total Liabilities**		**11,000**
Prepaid License Fees	**500**	Common Stock	40,000	
Machinery, net	4,800	Retained Earnings	(25)	
Trademark	5,000	**Total Owners' Equity**		**39,975**
Total Assets	**$50,975**	**Total Liabilities and Owners Equity**		**$50,975**

Changes in Non-Cash Current Assets and Liabilities

Note that all Assets, Liabilities and Shareholder's Equity accounts were zero at the beginning of the period.

DNA Storage, Inc.: CFO - Indirect Method

- **Items included in Net Income that never have a cash flow effect**
  - Add Depreciation (noncash expenses): **1,200**
- **Adjust for changes in operating (current) assets and liabilities**
  - **Subtract increase in current assets:**
    - Increase in Receivables: **3,500**
    - Increase in Materials: **2,300**
    - Increase in Prepaid License Fee: **500**
  - **Add increase in current liabilities:**
    - Increase in Accounts Payable: **4,000**
- **Subtract (add back) Non-operating gains (losses) included in NI**

DNA Storage, Inc.: CFO - Indirect Method

Net Income		**($25)**
Adjustments for non-cash expenses:		
Depreciation and amortization	$1,200	
Change in current assets and liabilities:		
Increase in Accounts receivable	(3,500)	
Increase in Pre-paid license fee	(500)	
Increase in Materials	(2,300)	
Increase in Accounts payable	4,000	
Total Adjustments		**(1,100)**
Cash Flows from (used in) Operations		**($1,125)**

DNA Storage, Inc.: Statement of Cash Flows 2017 (continued)

Cash from Investing:	
Acquisition of machinery	$(6,000)
Net cash from (used in) investing	**$(6,000)**

Cash from Financing:	
Issuance of common stock	$35,000
Bank borrowing	7,000
Net cash from (used in) financing	**$42,000**

Net Cash Flows	**$34,875**
Cash balance at the beginning of the year	**0**
Cash balance at the end of the year	**$34,875**

Note: CFI and CFF parts are the same as in the direct method